

BLOCK CIPHER MODES

Cryptography

CRYPTOGRAPHIC ALGORITHMS - BLOCK CIPHER MODES

A block cipher mode of operation specifies how block ciphers should handle the blocks. It uses a symmetric key block cipher algorithm to provide an information service. The service could be **authentication mode of operation** that provides a credentialing service or **unauthentication mode of operation** that provides a service such as confidentiality.

Electronic Code Book (ECB)

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Output Feedback Mode (OFB)

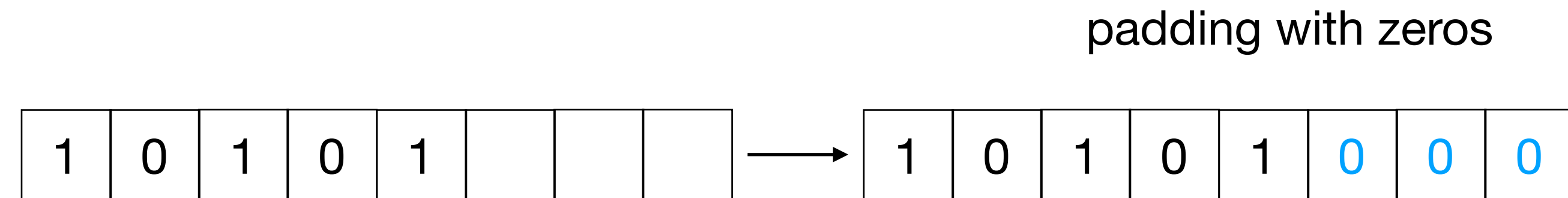
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Cryptography

CRYPTOGRAPHIC ALGORITHMS

128 bits, 192 bits, or 256 bits



Padding: Padding is a technique used to adjust the size of the plaintext to match the block size of the block cipher being used. In many encryption algorithms, data must be processed in fixed-size blocks, typically 128, 192, or 256 bits (16, 24, or 32 bytes). If the plaintext is not an exact multiple of the block size, padding is added to fill the last block.

Example: **Zero Padding:** In zero padding, zeros are added to the plaintext to fill the last block.

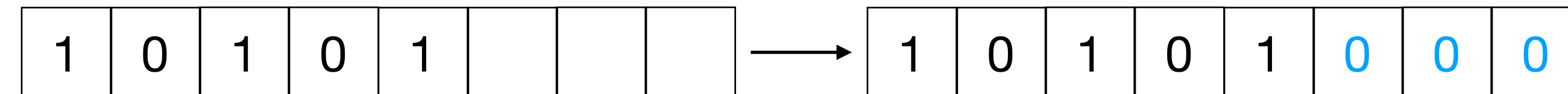
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padding with zeros



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Initialisation Vector (IV): The IV is a random or unique value used in various cryptographic processes, particularly in block cipher modes. Its purpose is to introduce randomness and ensure that the same plaintext, when encrypted multiple times with the same key, produces different ciphertext. This helps prevent patterns in the encrypted data, which can be exploited by attackers.

In the context of block cipher modes like CBC (Cipher Block Chaining), CFB (Cipher Feedback), OFB (Output Feedback), and CTR (Counter), the IV is used to XOR with the first block of plaintext before encryption. The IV should be unique for each message or session.

The use of the IV is particularly crucial in preventing known-plaintext attacks and replay attacks. For example, in CBC mode, using the same IV for different messages with the same key would result in the first block of ciphertext being identical, which can leak information about the common plaintext.

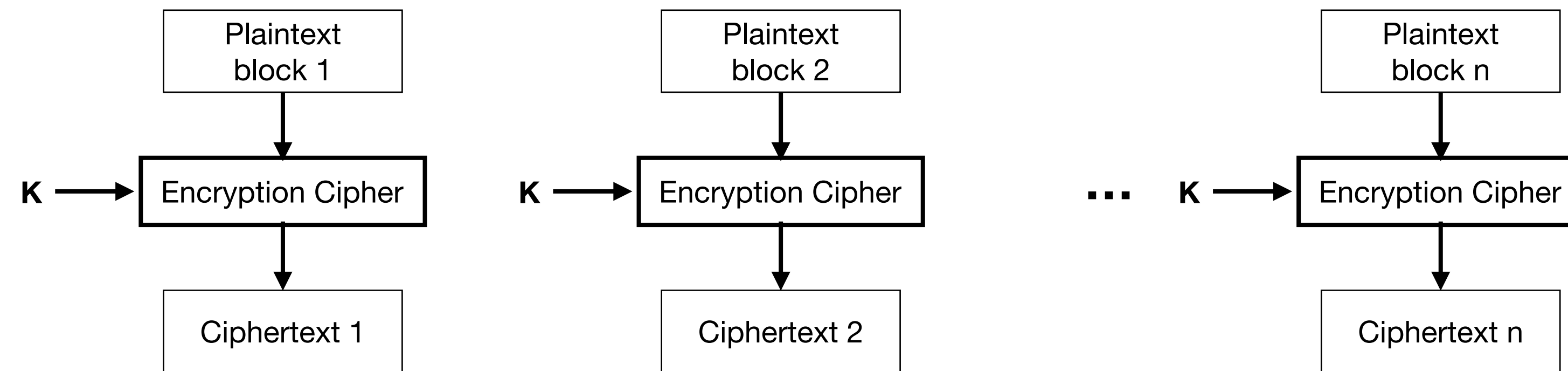
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CRYPTOGRAPHIC ALGORITHMS - BLOCK CIPHER MODES

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ENCRYPTION



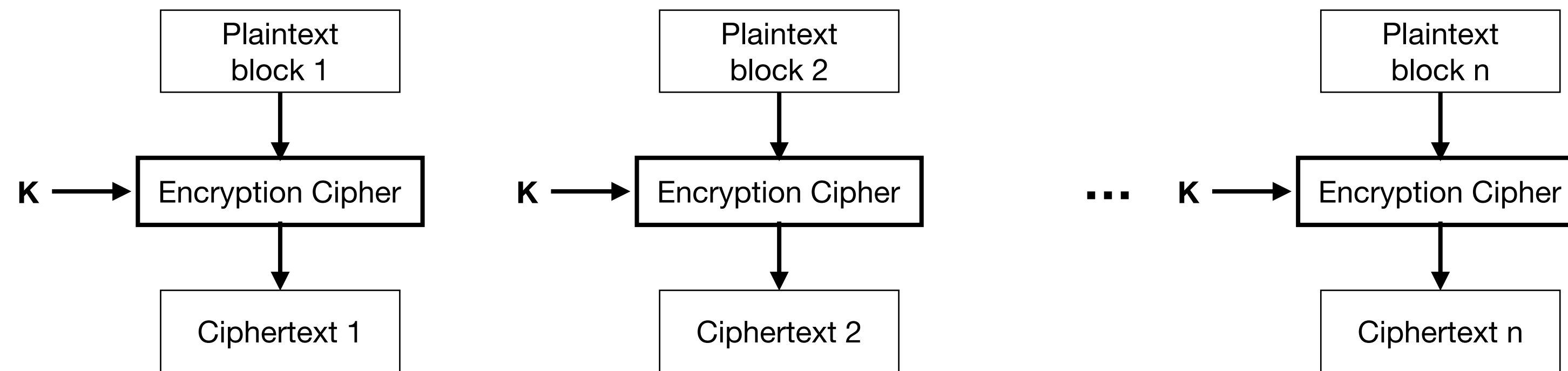
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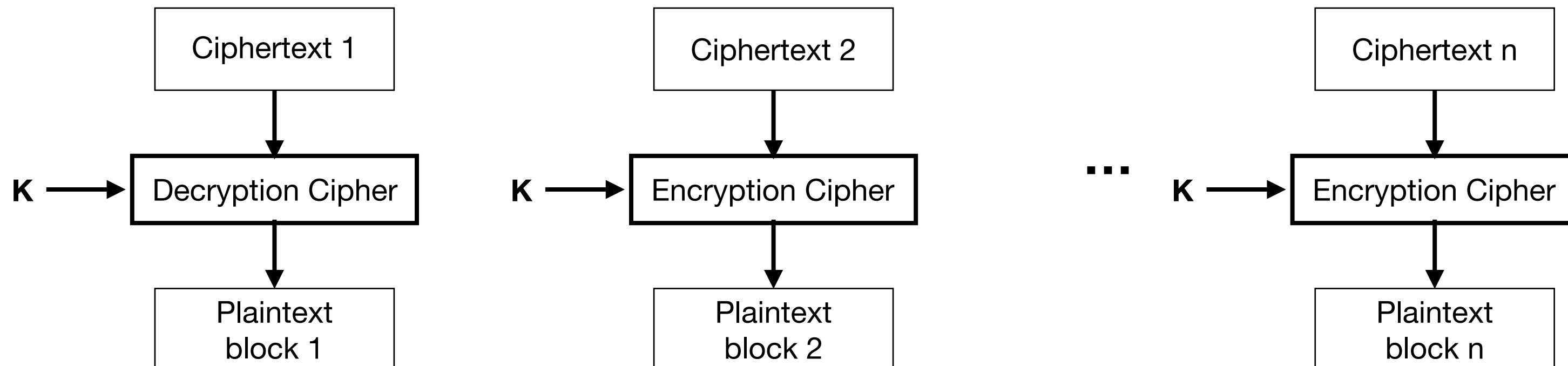
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DECRYPTION



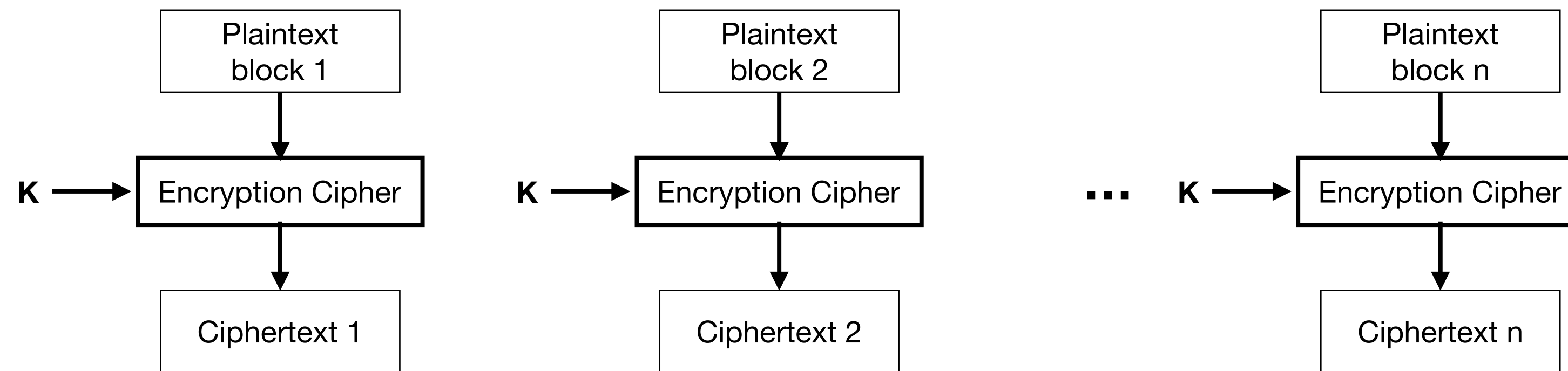
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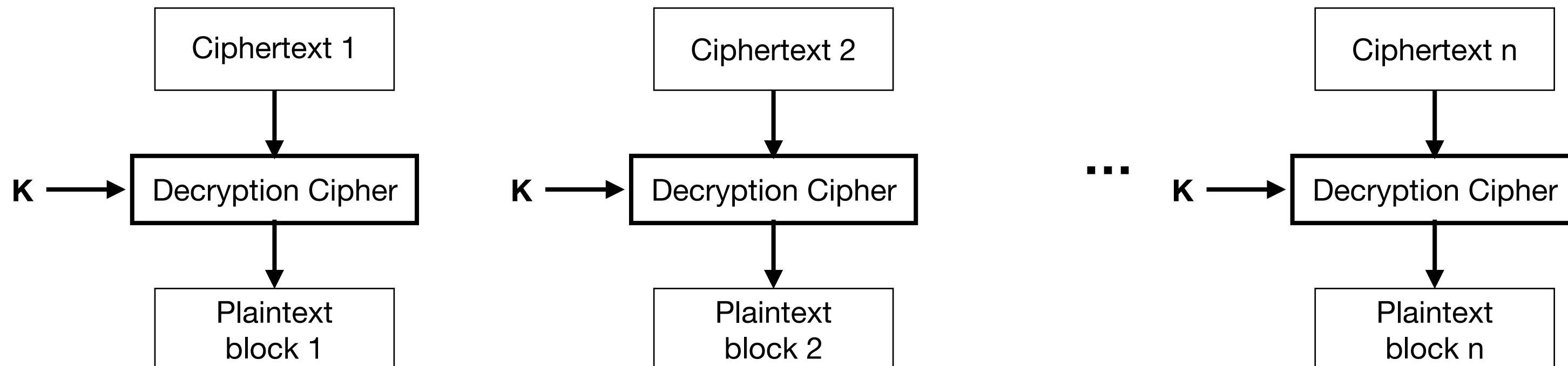


ADVANTAGES

- Parallel encryption of blocks of bits is possible, thus it is a faster way of encryption.
- Simple way of the block cipher.

Example: Encrypting product serial numbers

DECRYPTION



PURPOSE

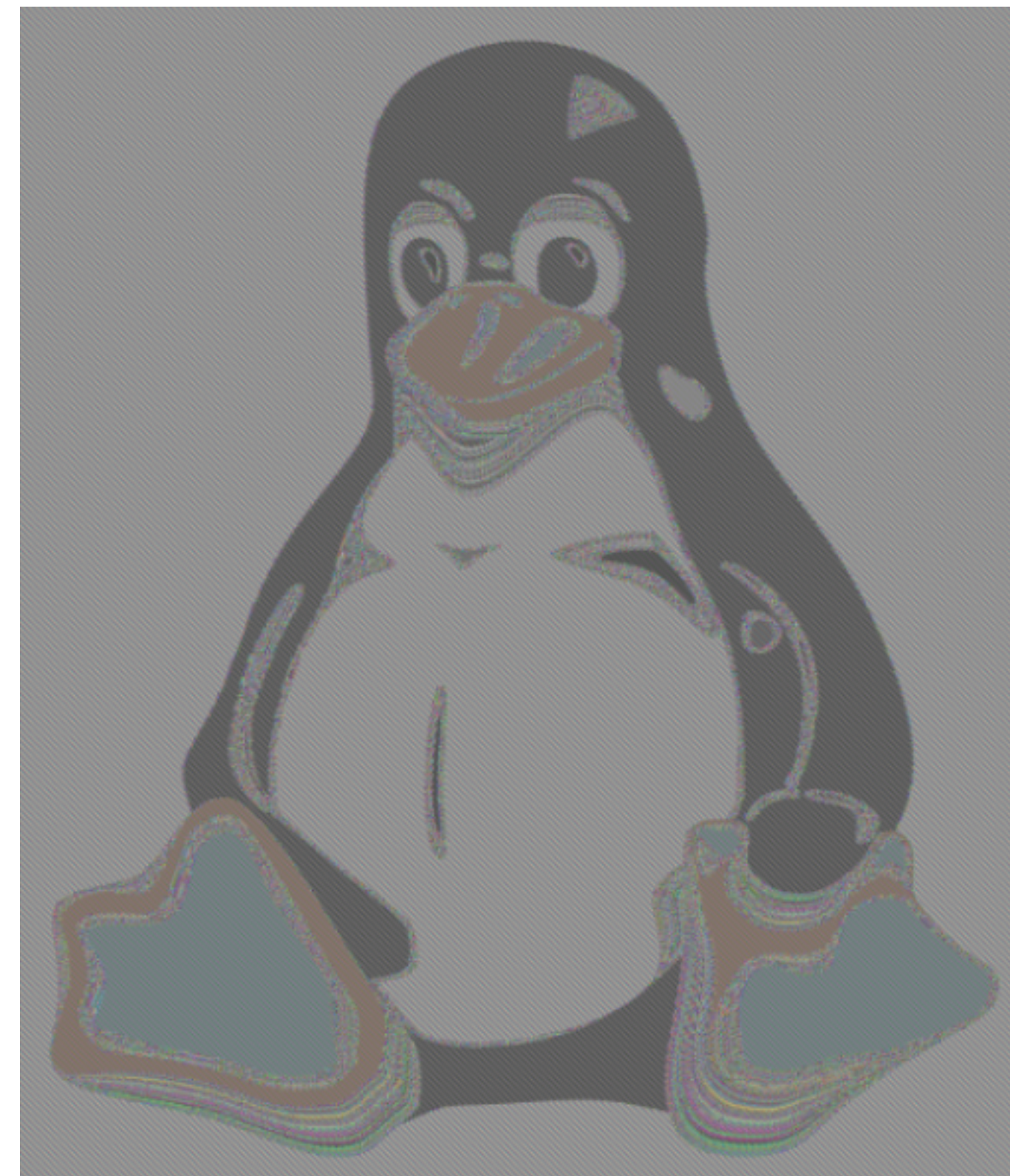
- Used when data blocks are independent and no synchronisation is required.
- ECB should not be used for sensitive data due to its vulnerabilities, including pattern leakage.

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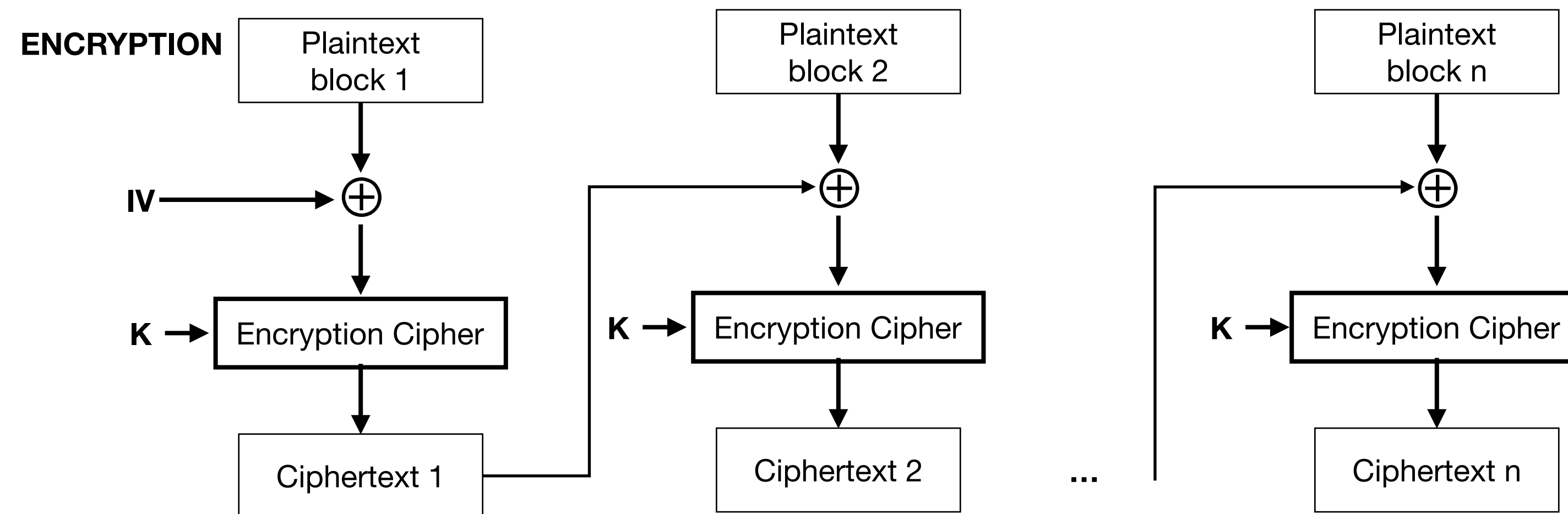


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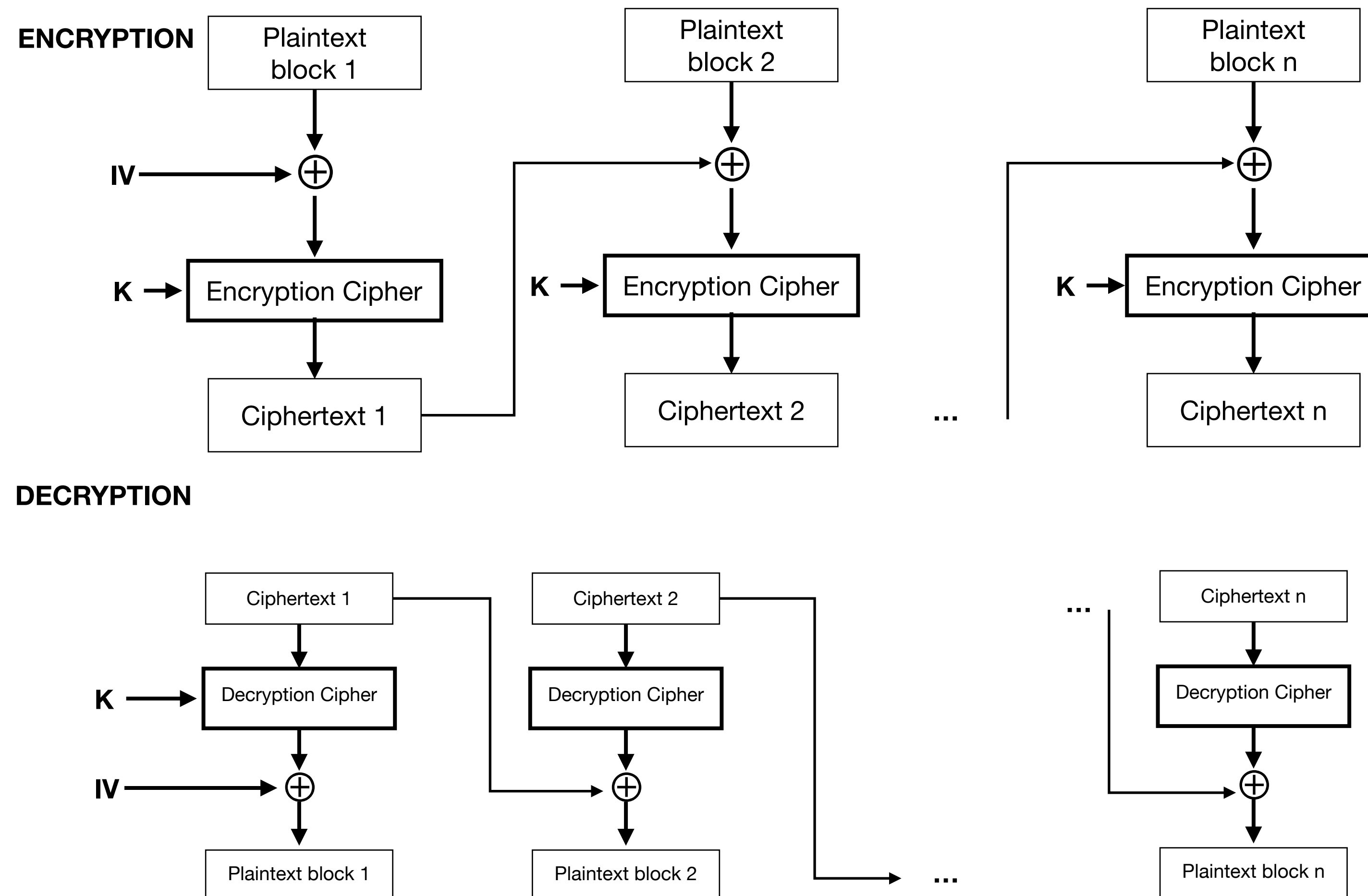


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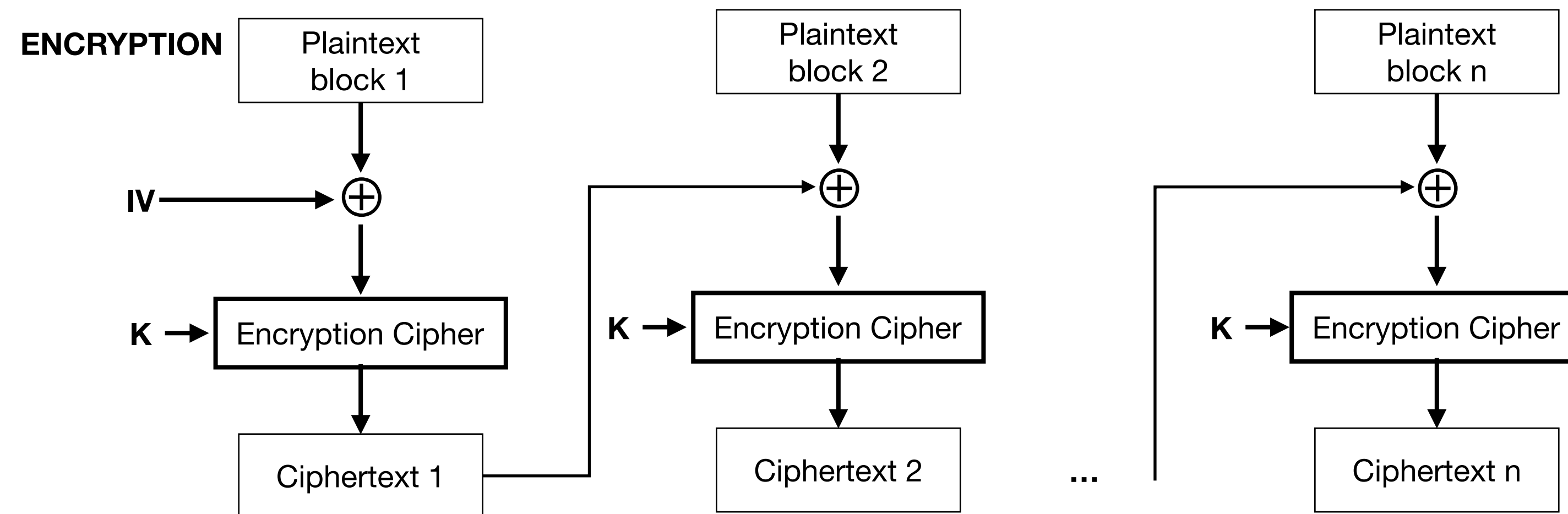


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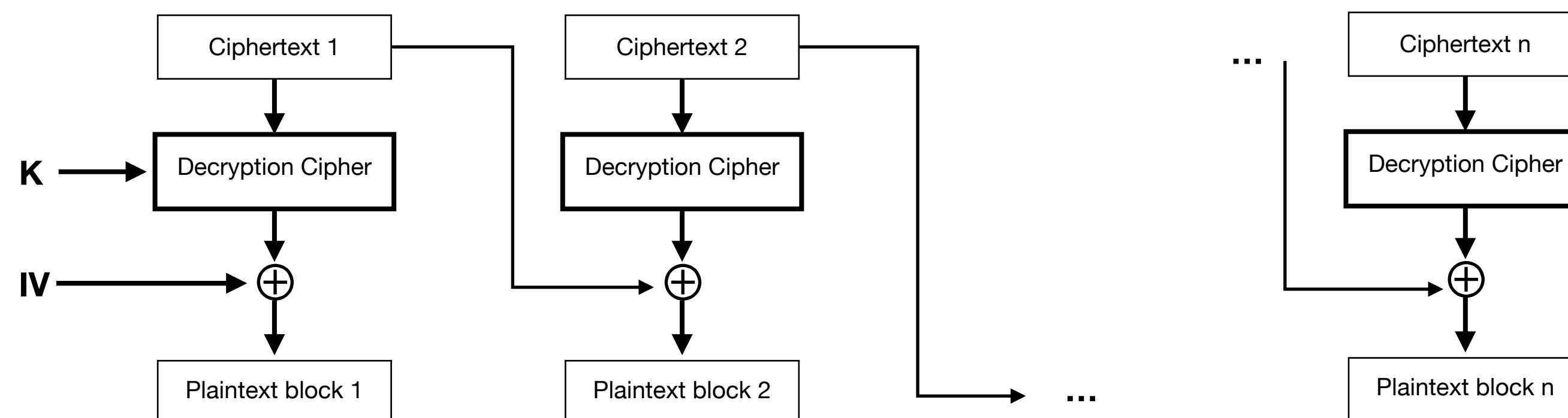
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ADVANTAGES

- Since, CBC works well for input greater than b bits.
- CBC is a good authentication mechanism.
- Better resistive nature towards cryptanalysis than ECB.

Examples: Disk encryption systems

PURPOSE

- CBC provides strong confidentiality and is suitable for securing data where data integrity is not critical.

Cryptography

CRYPTOGRAPHIC ALGORITHMS - BLOCK CIPHER MODES

Electronic Code Book (ECB) VS Cipher Blocking Chaining (CBC)

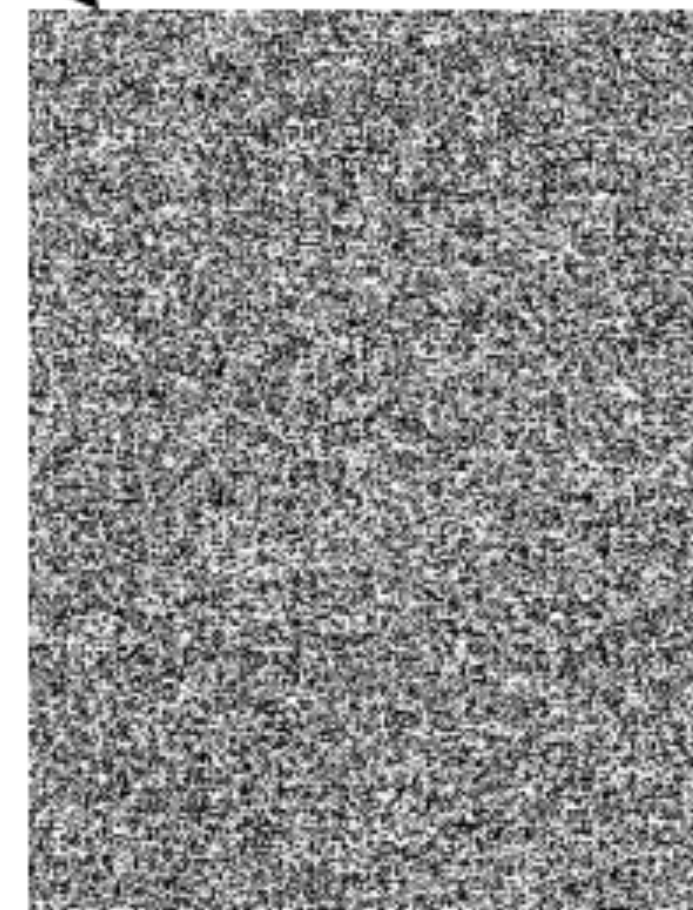
AES in ECB mode



AES in CBC mode



Similar plaintext
blocks produce
similar ciphertext
blocks (**not good!**)



Cryptography

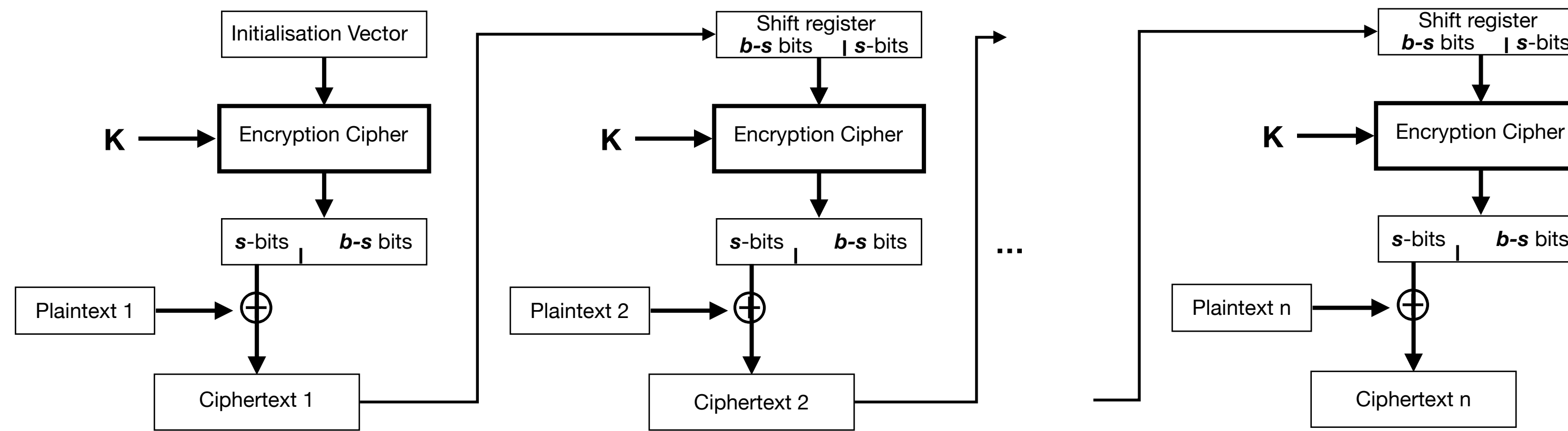
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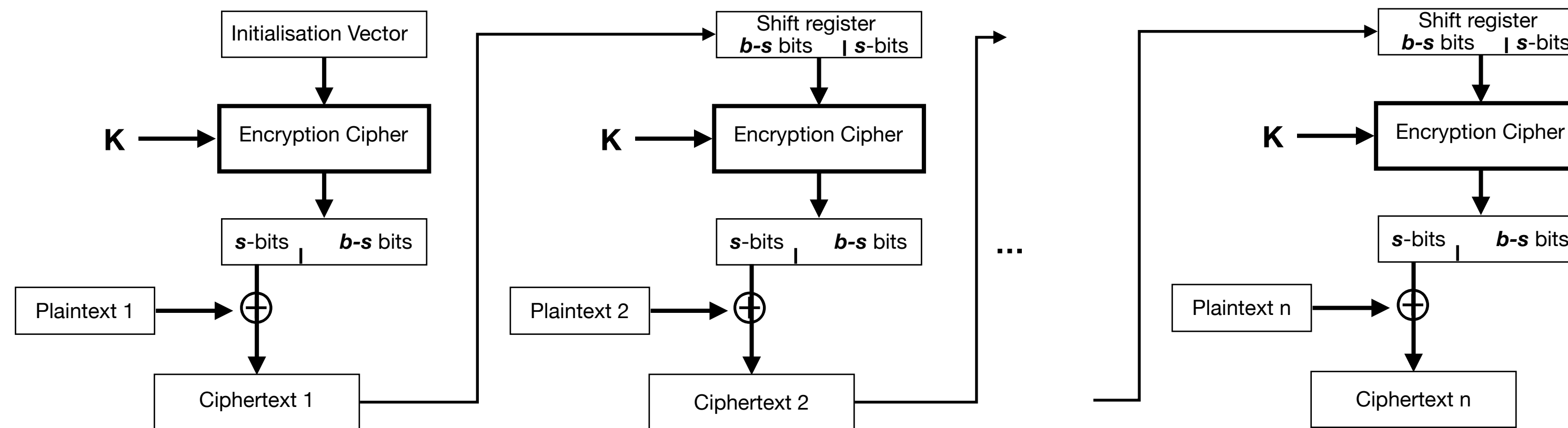
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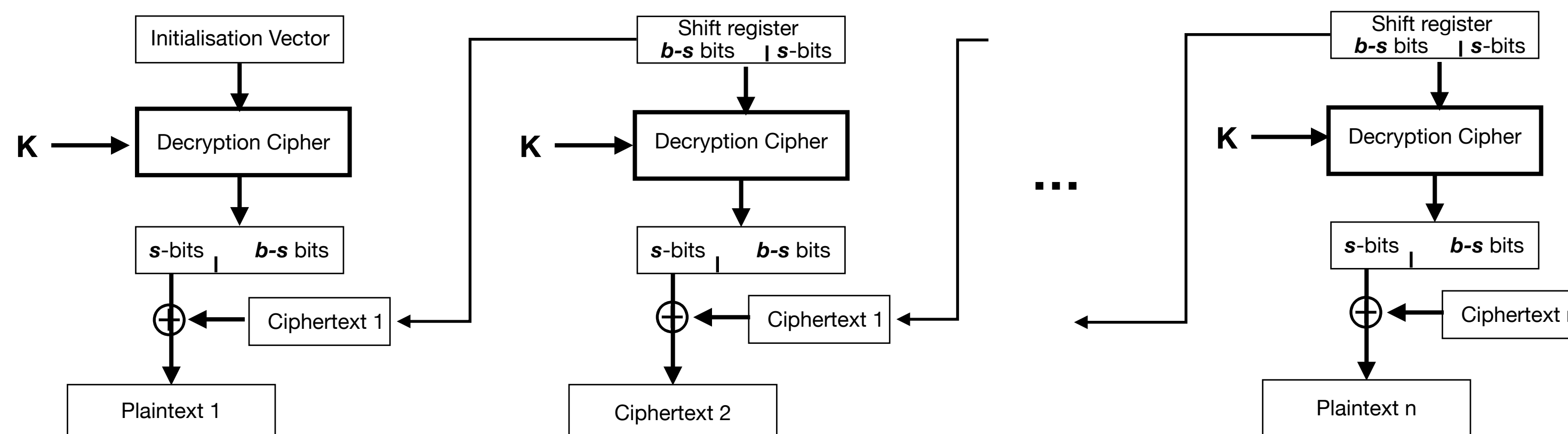
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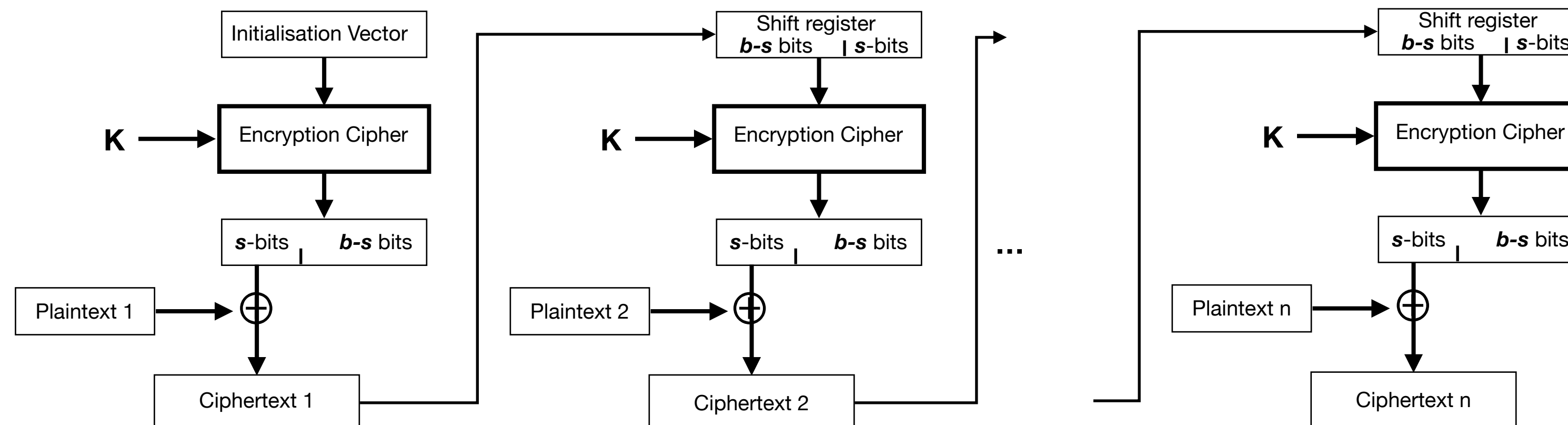
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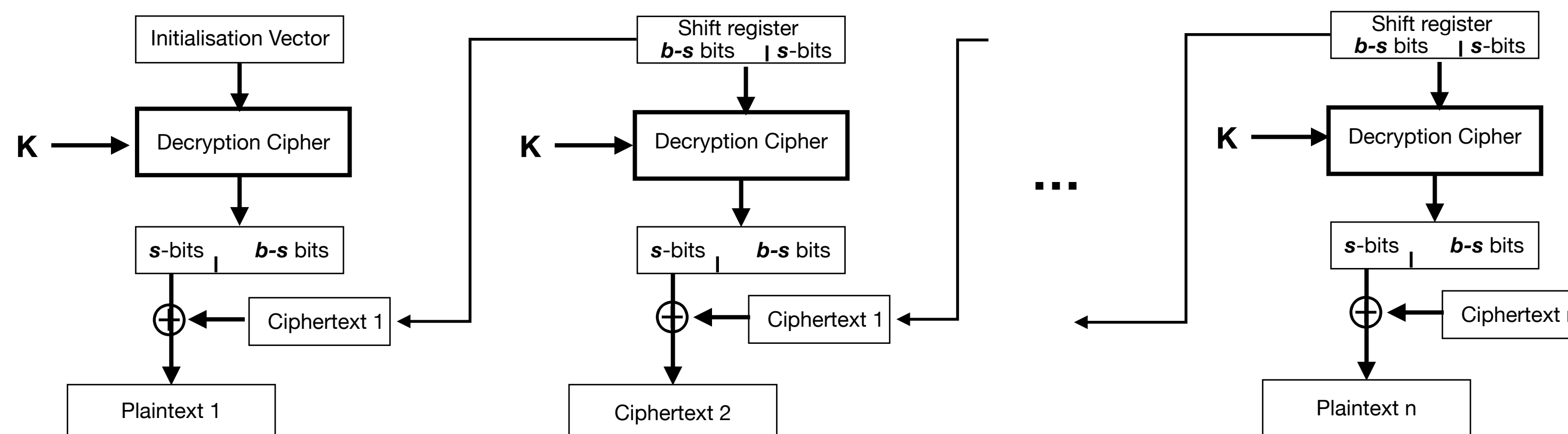


ADVANTAGES

- Since, there is some data loss due to the use of shift register, thus it is difficult for applying cryptanalysis.

Examples: real-time video streaming encryption

DECRYPTION



PURPOSE

- CFB is suitable when error propagation is not desired, and partial data corruption is tolerable.

Cryptography

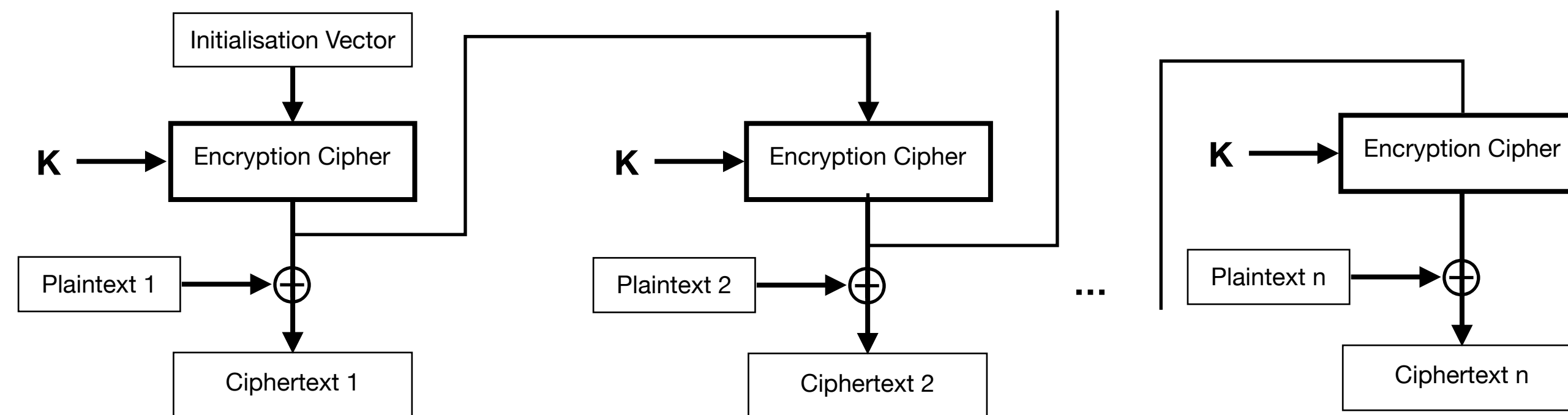
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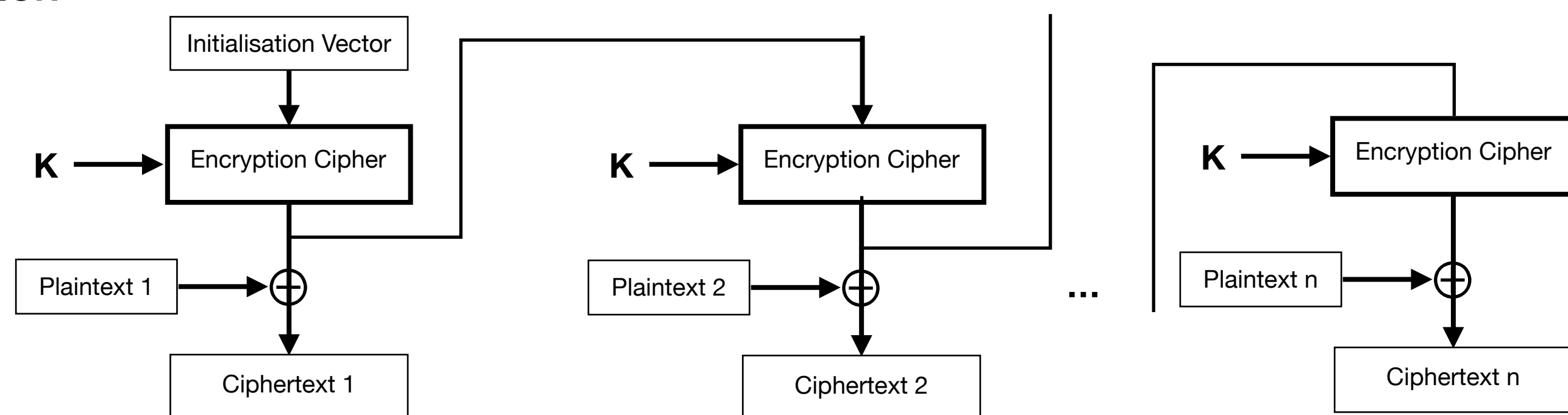
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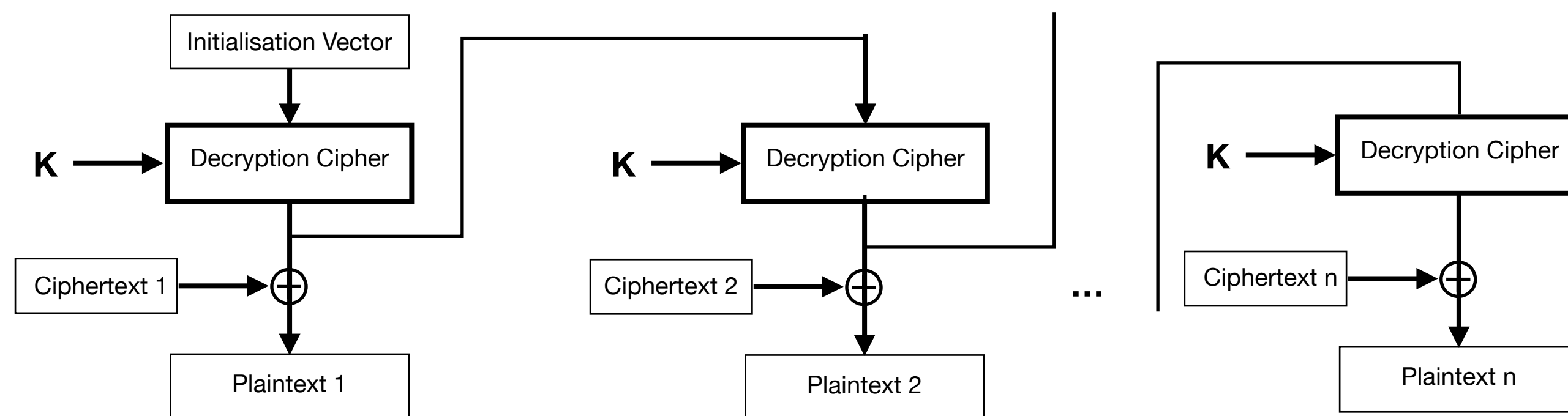
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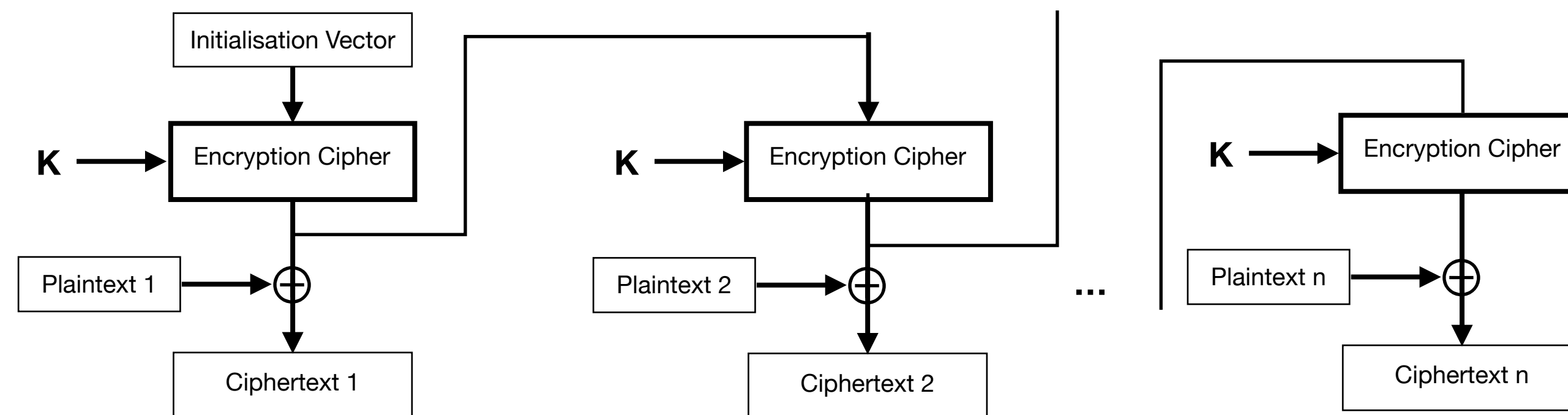
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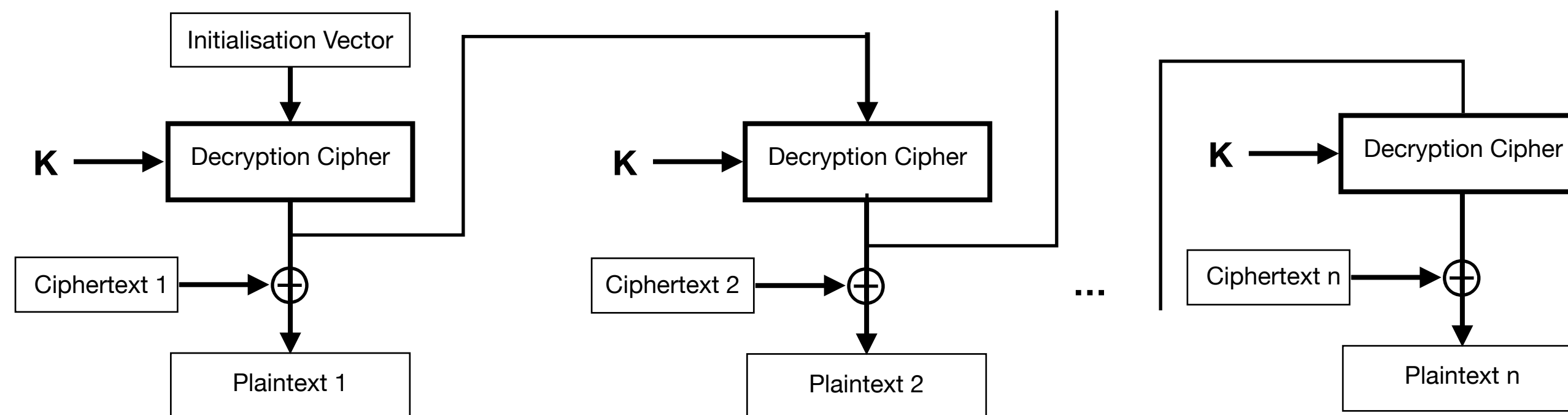
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ADVANTAGES

- In the case of CFB, a single bit error in a block is propagated to all subsequent blocks. This problem is solved by OFB as it is free from bit errors in the plaintext block.

Examples: Secure Remote Desktop Sessions

PURPOSE

- OFB is primarily used for error-tolerant applications and when encryption and decryption processes should be reversible.

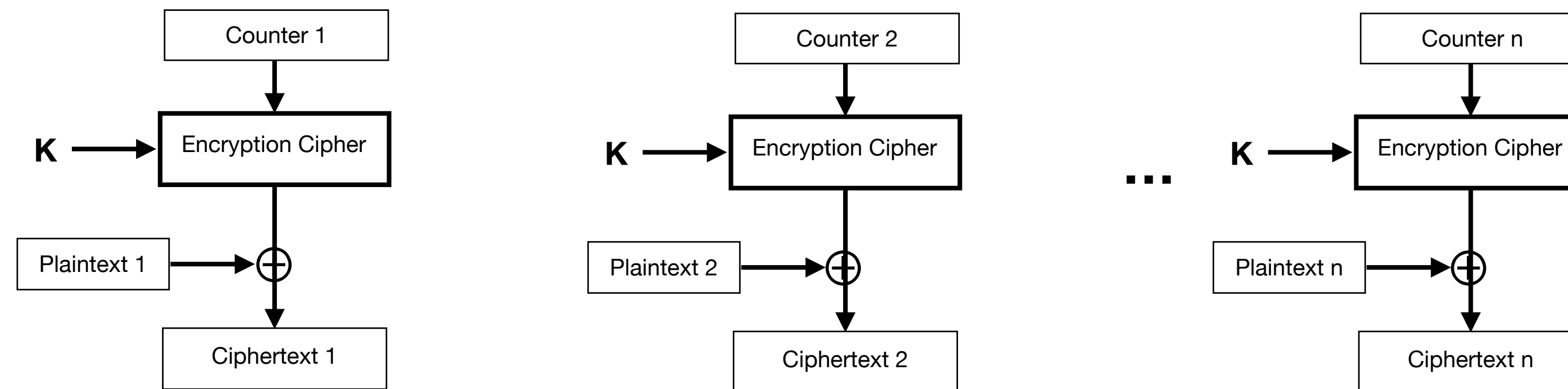
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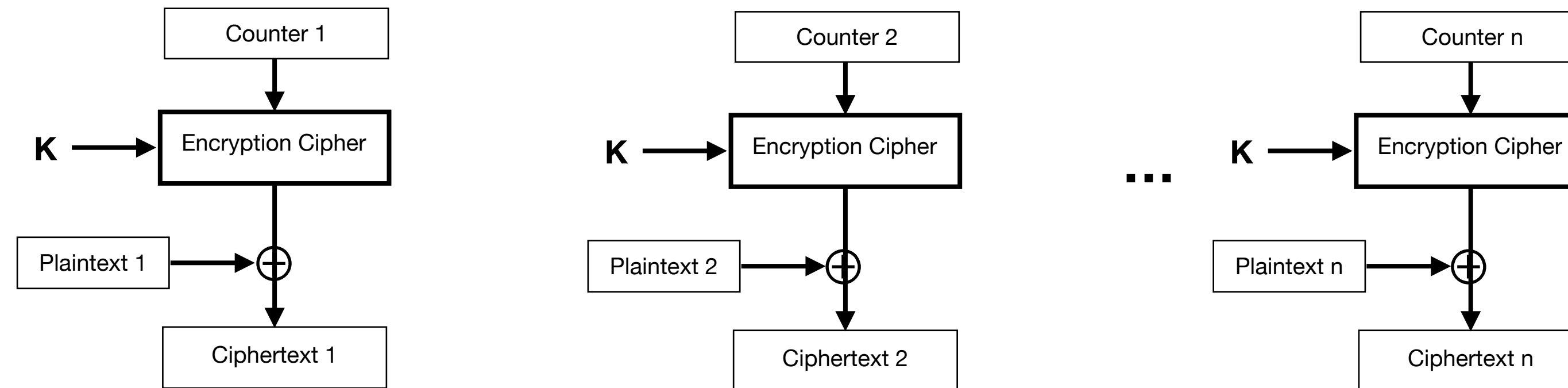
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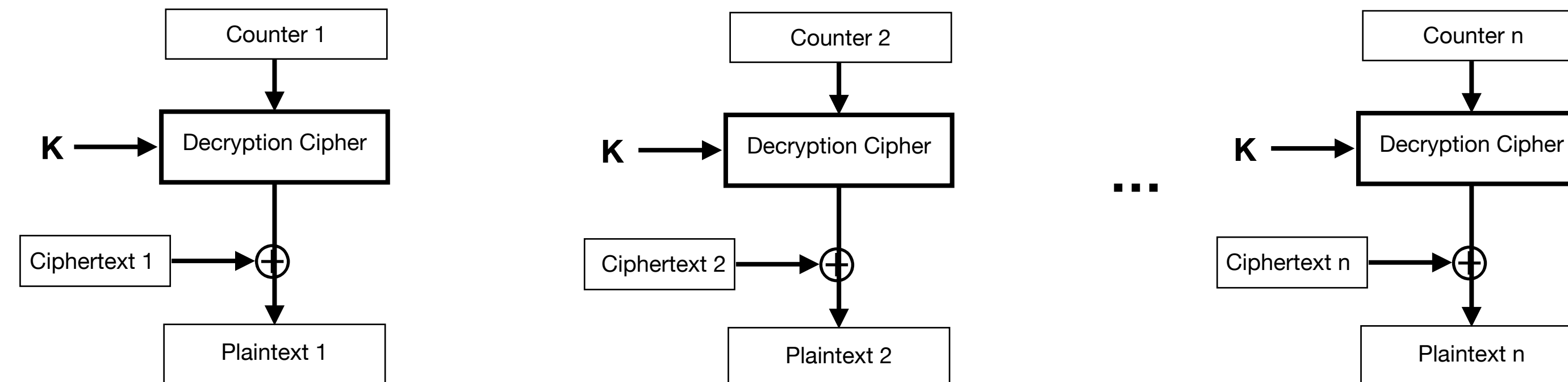
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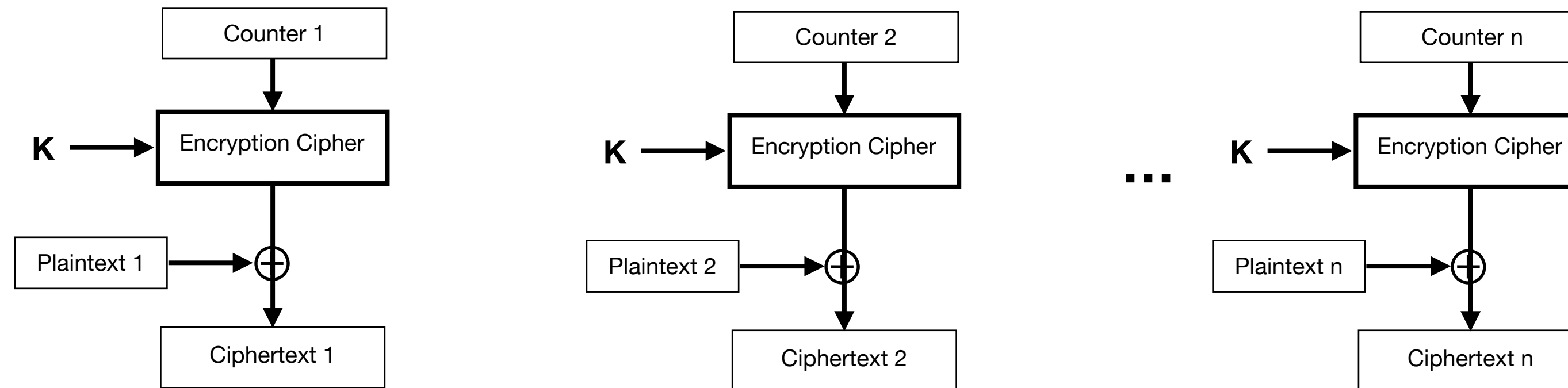
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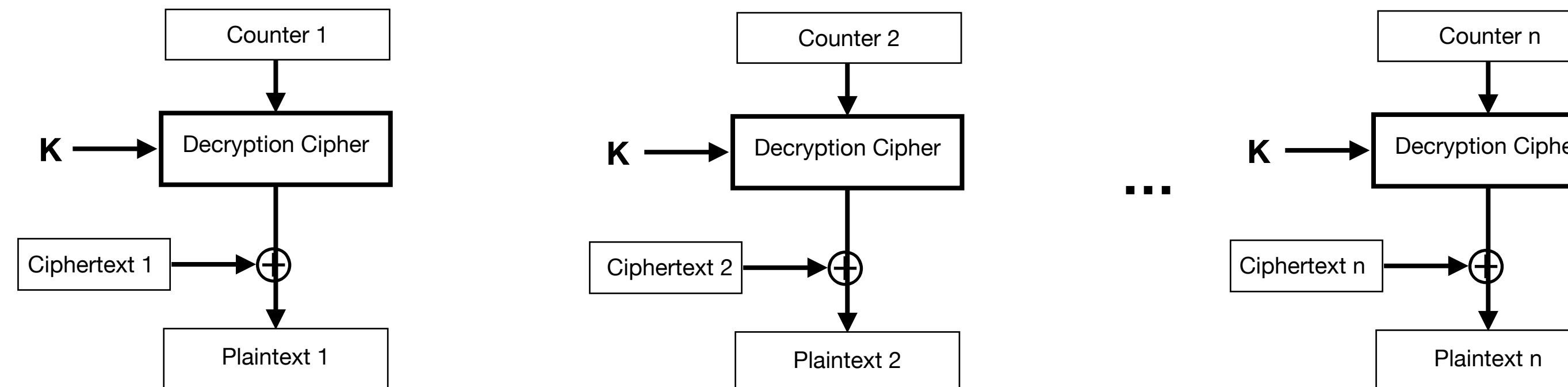
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ADVANTAGES

- Since there is a different counter value for each block, the direct plaintext and ciphertext relationship is avoided. This means that the same plain text can map to different ciphertext.
- Parallel execution of encryption is possible as outputs from previous stages are not chained as in the case of CBC.

Examples: Secure Cloud Storage

PURPOSE

- CTR provides strong security, parallelisation, and is suitable for applications where encryption and decryption must be reversible and where error propagation should be avoided.

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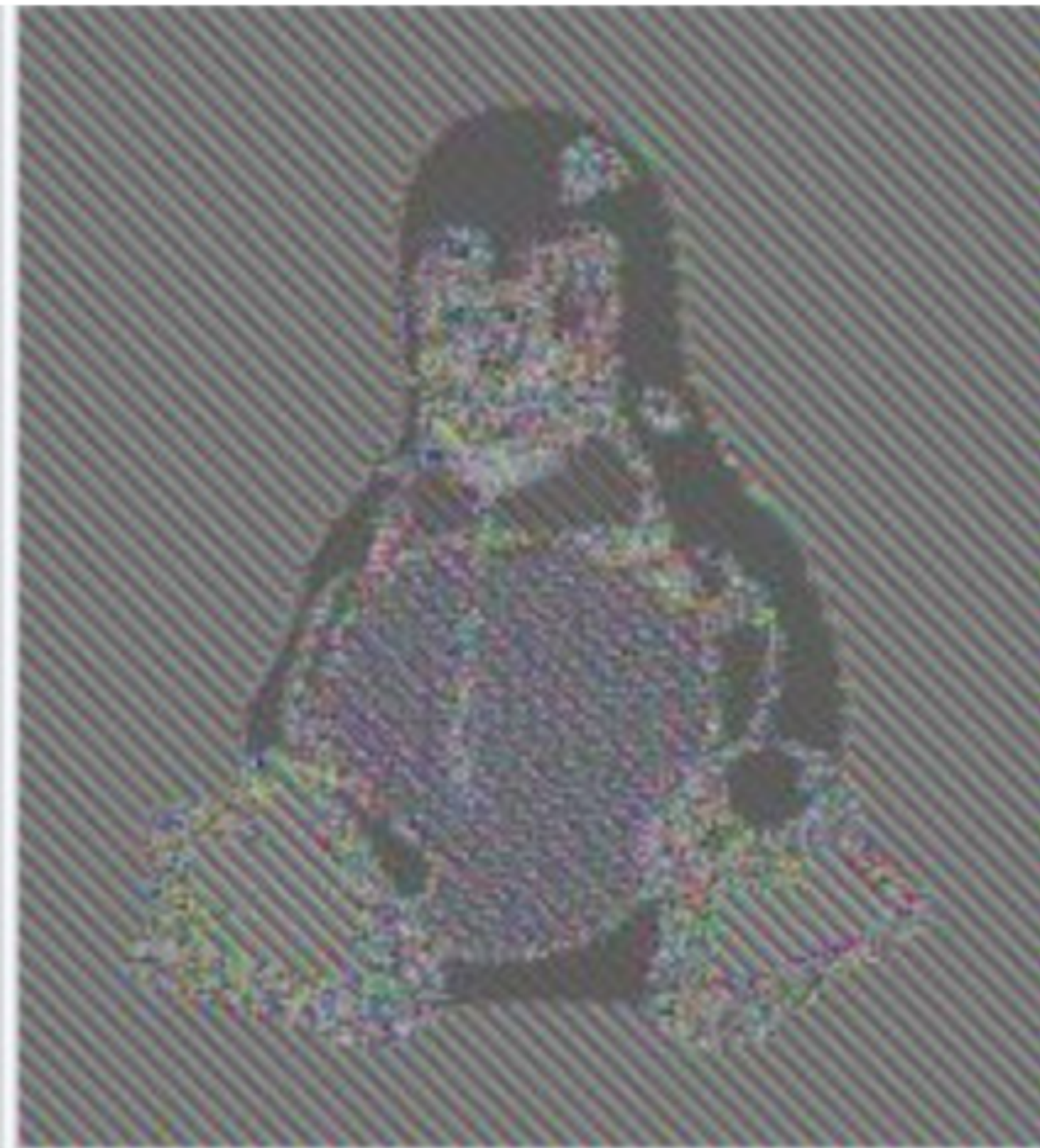
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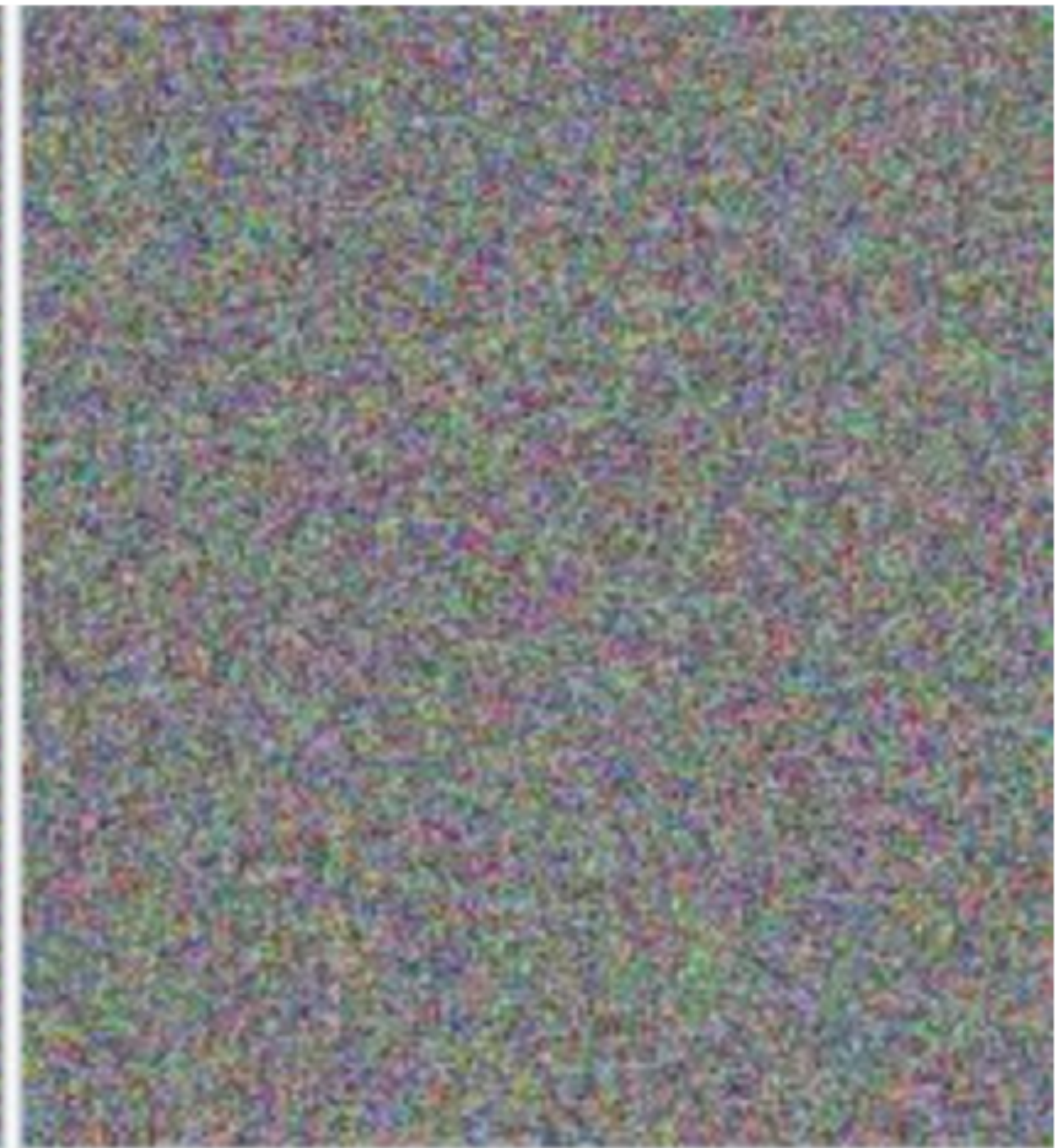
Plaintext



ECB



CTR

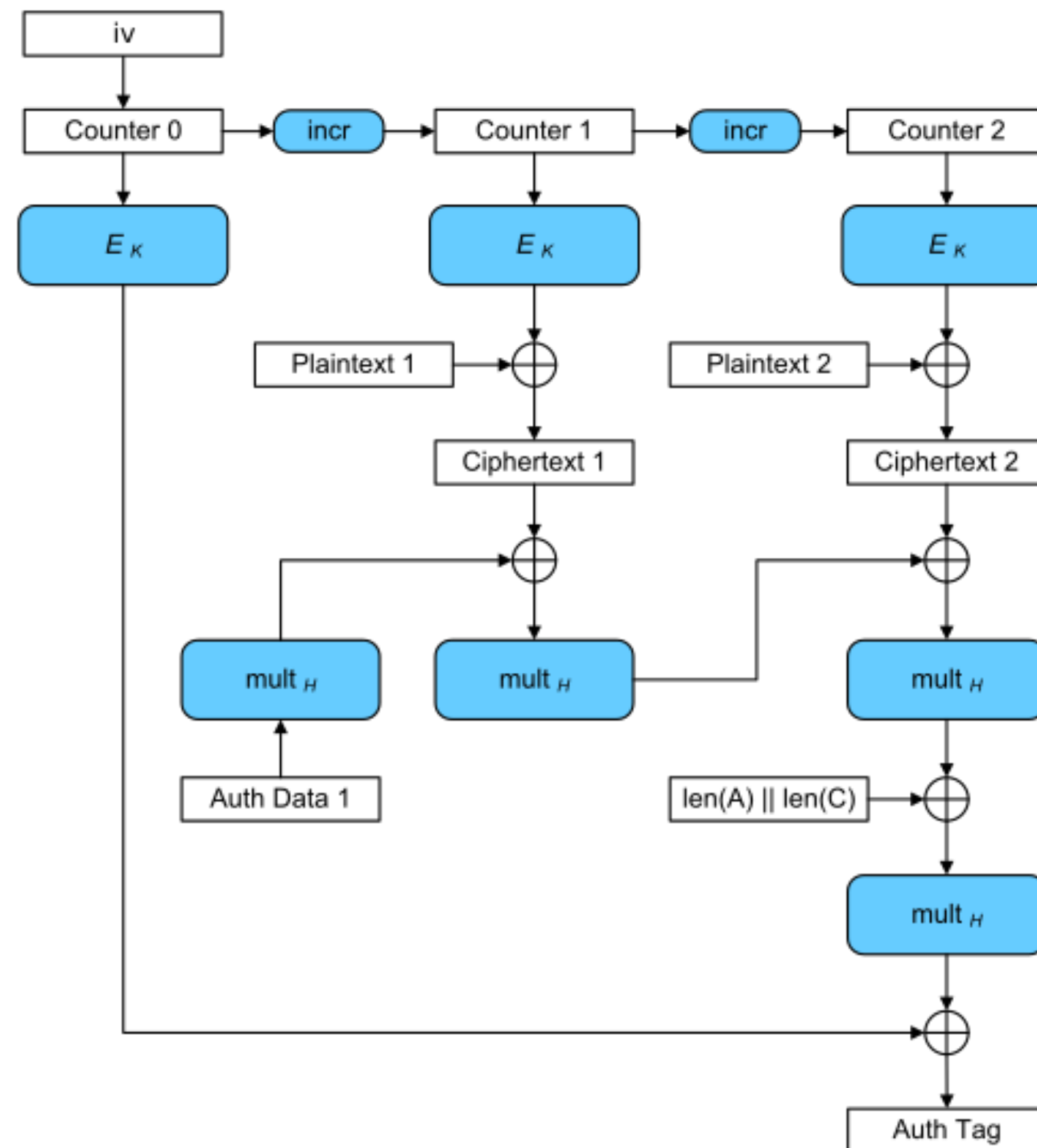


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AES - Galois/Counter Mode (GCM)

AES-GCM is a widely used block cipher mode that combines the security of the AES block cipher with the Galois/Counter Mode (GCM) for efficient encryption and authentication. It provides both confidentiality and data integrity, making it suitable for secure communication over untrusted networks. AES-GCM uses a unique counter for each block of data, ensuring encryption is parallelizable and resistant to replay attacks.



ADVANTAGES

- Parallel processing
- Resistance to Replay Attacks: